

UpperclassAI complete, structured digital learning curriculum for The AI tutor. It is organized into 4 progressive levels—from a foundation in AI literacy to practical application and advanced technical skills.

The curriculum is designed to work with The platform's adaptive engine: the AI tutor should use knowledge tracing to assess a student's existing knowledge in a domain before advancing them to more complex topics, and it should generate personalized explanations and real-world examples based on the student's interests.

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## Level 1: Foundations of AI & Digital Literacy (Beginner)

Goal: To demystify AI, establish core technical vocabulary, and introduce the concept of algorithms as the basis for computational thinking. This stage focuses on "Learning about AI".

- Module 1.1: What is Artificial Intelligence?
  - Topic 1: Defining AI and differentiating it from automation and human intelligence.
  - Topic 2: A brief history of AI: From the Turing Test to Generative AI.
  - Topic 3: Real-world applications of AI in daily life (recommendation engines, facial recognition, virtual assistants).
- Module 1.2: The Building Blocks of AI
  - Topic 1: What are Algorithms? (Tutor Activity: The AI tutor can demonstrate an algorithm by guiding the student through a simple "sorting" or "pattern-finding" game).
  - Topic 2: Introduction to Data: How AI learns from information (structured vs. unstructured data).
  - Topic 3: What is Machine Learning (ML)? A simple breakdown of supervised, unsupervised, and reinforcement learning.

- Module 1.3: Effective Human-AI Collaboration (Prompting 101)
  - Topic 1: How to communicate with an AI: Understanding Large Language Models (LLMs).
  - Topic 2: The Art of Prompting: Writing clear, contextual instructions.
  - Topic 3: Exercise: Iterative prompting (student asks the AI tutor to refine a response to make it simpler, more creative, or more factual).

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## Level 2: Applied AI & Responsible Use (Intermediate)

Goal: To shift from theory to practice. The student learns how to use AI tools for productivity and creativity while developing a critical eye for bias, ethics, and misinformation. This stage focuses on "Learning with AI".

- Module 2.1: Generative AI for Creation
  - Topic 1: Text Generation: Using AI for summarization, brainstorming, and drafting (Tutor Activity: The AI helps the student outline a blog post or a study guide).
  - Topic 2: Image & Media Generation: How text-to-image models work.
  - Topic 3: Ethics of Generation: Ownership, copyright, and the difference between human-made and AI-generated content.
- Module 2.2: AI for Data & Analysis
  - Topic 1: Introduction to Data Analytics for decision-making.
  - Topic 2: Using AI to interpret charts, spreadsheets, and reports.
  - Topic 3: Predictive AI in business and healthcare (case studies).
- Module 2.3: Navigating the Challenges (AI Safety)
  - Topic 1: Bias in AI: How training data affects outcomes.
  - Topic 2: Hallucinations and Misinformation: Why AI "makes things up" and how to fact-check using the SIFT method (Stop, Investigate, Find, Trace).

- Topic 3: Privacy and Security: Data sharing risks.

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### Level 3: Technical Implementation (Advanced)

Goal: To equip the learner with the hard skills required to build and customize AI tools. This stage is ideal for students who wish to pursue careers in tech.

- Module 3.1: Programming Foundations for AI
  - Topic 1: Introduction to Python (Variables, data types, loops, functions) using Jupyter Notebooks.
  - Topic 2: Working with Data using Pandas.
  - Topic 3: Data Visualization using Matplotlib.
- Module 3.2: Building with Machine Learning
  - Topic 1: Introduction to Scikit-learn: Building a simple classification model.
  - Topic 2: Understanding Neural Networks.
  - Topic 3: Natural Language Processing (NLP): Sentiment analysis and text classification.
- Module 3.3: Creating Custom AI Assistants
  - Topic 1: Understanding Retrieval-Augmented Generation (RAG).
  - Topic 2: How to build a Custom GPT or Assistant (Role, Constraints, Knowledge).
  - Topic 3: Project: The student uses the AI tutor's backend tools to design a "specialist tutor" (e.g., a "Math Mentor" or "Code Debugger").

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### Level 4: The Future & Synthesis (Expert/Project-based)

Goal: To synthesize all previous knowledge to solve real-world problems and debate the future societal impact of AI.

- Module 4.1: AI in the Workforce
  - Topic 1: AI and the Future of Work: Automation vs. Augmentation.
  - Topic 2: Building a professional brand with AI (Optimizing LinkedIn, generating portfolios).
  - Topic 3: AI Project Management: How to lead teams that use AI tools.
- Module 4.2: The Ethics of Autonomous Systems
  - Topic 1: Deepfakes and Digital Literacy: Identifying manipulated media.
  - Topic 2: AI for Social Good vs. Surveillance.
  - Topic 3: Debate simulation: The student argues a position on AI regulation with the AI tutor acting as a Devil's Advocate.
- Module 4.3: Capstone Project (The Tutor's Exam)
  - Task: The student must identify a problem (e.g., "I need to learn Spanish verbs" or "My small business needs a marketing plan") and use the AI tools learned in the course (Prompting, Python, Custom Assistants) to solve it.
  - Assessment: The AI tutor evaluates the student's process, the accuracy of the output, and the student's reflective analysis on the limitations of their solution.

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## How The AI Tutor should Use This Curriculum

To make this work, The backend logic (Django + ML) needs to connect to this content in three specific ways:

### 1. Knowledge Tracing (The "Check for Understanding")

- As the user interacts with the React frontend, the ML model should map their responses to specific Skills (e.g., Skill\_ID: PROMPT\_101).

- If a student fails Module 1.3 (Prompting), the model should prevent them from jumping to Module 2.1 (GenAI Creation) until they have reviewed and passed a remedial quiz. This ensures "Curriculum Alignment".

## 2. Adaptive Explanations (The "Human" Touch)

- Don't just show text from a database. Use an LLM (via API) to re-explain concepts based on user data.

- Example: If the user is a gamer and asks about "Algorithms," the prompt engineering layer should instruct the LLM to respond with: "Think of an algorithm like a speedrun strategy guide. It's a step-by-step sequence to solve a level (problem) efficiently."

## 3. Scaffolding Assessments

- Use the Generative AI capabilities to create unique assessments on the fly. If a student struggles with "Bias in AI," the tutor should generate 3 new "real-world scenario" questions dynamically, rather than repeating the same failed quiz question.

This curriculum is designed to be iterative. Start with Level 1 fully coded into the database, monitor which lessons have the highest drop-off rates using the analytics dashboard, and let the AI learn which teaching methods work best for subsequent updates.